INCH-POUND
MIL-DTL-32708A
30 January 2024
SUPERSEDING
MIL-DTL-32708
17 May 2022

DETAIL SPECIFICATION

LINER, TRIGGER FINGER MITTEN, EXTREME COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the extreme cold weather trigger finger mitten liner. This item is designed for wear in combination with a full mitten shell or a trigger finger mitten shell conforming to types I and II of MIL-DTL-32707.

1.2 Classification. The mitten liner is of one type in the following sizes, as specified (see 6.2).

1.2.1 Sizes.

Small

Medium

Large

Extra Large

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59826 - Thread, Nylon

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-C-3735 - Cuffs, Knit, Wrist and Ankle, and Cloth, Knitted

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8440

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MIL-PRF-32567	-	Batting, Synthetic Fibers, Unquilted
MIL-DTL-32697	-	Cloth, Napped French Terry
MIL-DTL-32699	-	Cloth, 70 Denier Nylon Ripstop
MIL-DTL-32707	-	Mitten Shells, Full and Trigger Finger, Extreme Cold Weather

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

United States Standards for Grades of Wool

(Copies of this document are available online at https://www.ams.usda.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP1	-	Gray Scale for Color Change
AATCC EP9	-	Visual Assessment of Color Difference of Textiles
AATCC TM20	-	Test Method for Fiber Analysis: Qualitative
AATCC TM118	-	Oil Repellency: Hydrocarbon Resistance Test
AATCC TM127	-	Test Method for Water Resistance: Hydrostatic Pressure
AATCC TM135	-	Dimensional Changes of Fabrics After Home Laundering
AATCC TM150	-	Dimensional Changes of Garments after Home Laundering

(Copies of these documents are available online at www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ/ANSI Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D737	-	Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1610	-	Standard Practice for Conditioning Leather and Leather Products for Testing
ASTM D1777	-	Standard Test Method for Thickness of Textile Materials
ASTM D1814	-	Standard Test Method for Measuring Thickness of Leather Units
ASTM D1912	-	Standard Test Method for Cold-Crack Resistance of Upholstery Leather
ASTM D2209	-	Standard Test Method for Tensile Strength of Leather
ASTM D2582	-	Standard Test Method for Puncture-Propagation Tear Resistance of Plastic Film and Thin Sheeting
ASTM D2617	-	Standard Test Method for Total Ash in Leather

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ASTM D2807	-	Standard Test Method for Chromic Oxide in Leather (Perchloric Acid Oxidation)
ASTM D2810	-	Standard Test Method for pH of Leather
ASTM D2821	-	Standard Test Method for Measuring the Relative Stiffness of Leather by Means of a Torsional Wire Apparatus
ASTM D3495	-	Standard Test Method for Hexane Extraction of Leather
ASTM D3775	-	Standard Test Method for End (Warp) and Pick (Filling) Count of Woven Fabrics
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3884	-	Standard Guide for Abrasion Resistance of Textile Fabrics (Rotary Platform Abrader Method)
ASTM D4705	-	Standard Test Method for Stitch Tear of Leather, Double Hole
ASTM D5052	-	Standard Test Method for Permeability of Leather to Water Vapor
ASTM D6076	-	Standard Test Method for Shrinkage Temperature of Leather
ASTM D6182	-	Standard Test Method for Flexibility and Adhesion of Finish on Leather
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM D6797	-	Standard Test Method for Bursting Strength of Fabrics Constant-Rate-of-Extension (CRE) Ball Burst Test
ASTM D7255	-	Standard Test Method for Abrasion Resistance of Leather (Rotary Platform, Abraser Method)
ASTM E96/E96M	-	Standard Test Methods for Gravimetric Determination of Water Vapor Transmission Rate of Materials
ASTM F3426	-	Standard Test Method for Measuring the Thermal Insulation of Clothing Items Using Heated Manikin Body Forms

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 11640	-	Leather — Tests for Colour Fastness — Colour Fastness to Cycles of To-and-Fro Rubbing
ISO 17075-2	-	Leather — Chemical Determination of Chromium(VI) Content in Leather — Part 2: Chromatographic Method

(Copies of these documents are available online at www.iso.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

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3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Standard sample. The finished trigger finger mitten liner and design shall match the standard sample (see 6.4) for design and all other characteristics or acceptance criteria for which a standard sample is referenced.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Design. The trigger finger mitten liner is an inner mitten designed to be worn in combination with a mitten shell in extreme cold weather environments. The trigger finger design enables the wearer to move their first finger independently from the rest of the hand. The liner is part of a system that includes a waterproof full mitten shell, or trigger finger mitten shell conforming to type I and type II of MIL-DTL-32707. The liner can be worn over a glove insert if desired. The mitten liner consists of the following: the outside shell is constructed from a water repellant 70 denier nylon ripstop conforming to 3.5.1MIL-DTL-32699 and a water repellent goatskin leather trigger finger conforming to 3.5.2. Grip material conforming to 3.5.3 is sewn at the fingertips on the outside of the nylon ripstop. The mitten liner is insulated on the front and the back of the hand, and the outer thumb, with material conforming to 3.5.6. The inside cuff and trigger finger is lined with napped wool French terry conforming to 3.5.4; the lining used on the front of the hand, back of the hand, and the inner and outer thumb is a napped wool lay-in conforming to 3.5.5. Each liner has an attached cuff conforming to 3.5.7. The liner has a label to identify the item, its size, its use, and care.

3.4.1 Figures. Figures 1 and 2 are furnished for informational purposes only. To the extent an inconsistency exists between the written specification and the figures, the written specification shall govern.

3.5 Materials and components.

3.5.1 Liner shell material. The cloth used to construct the liner's outer shell shall be a water repellent urethane coated 70 denier nylon ripstop conforming to MIL-DTL-32699 (see 6.6.1). Unless otherwise specified (6.2), the color shall be Coyote 498 (see 6.5.2).

3.5.2 Leather. The leather used to construct the trigger finger shall be a 2.25- to 2.75-ounce full grain goatskin conforming to the requirements in table I when tested as specified in 4.4.3. The leather shall be chrome tanned, light finish, fungicide and durable water repellent treated, backed by a waterproofing layer. Unless otherwise specified (see 6.2), the shade of the leather shall be Coyote 498 (see 6.5.2) and shall be drum dyed, struck through from grain to flesh.

TABLE I. Leather characteristics and requirements.

Characteristic	Requirement
Thickness, ounces	2.25 to 2.75
Moisture vapor transmission rate (MVTR), grams per square meter in 24 hours, upright (minimum)	500
Water absorption resistance, percent absorption (maximum) 1/	15
Oil and stain resistance, grade (minimum)	4
Abrasion resistance at 1,000 cycles with a CS-10 wheel and 500-gram load	No russet exposure
Stiffness, degrees (maximum) 1/	50
Stitch tear, pounds (minimum) 1/	17
Elongation at 25-pound load, percent (maximum) 1/	25
Shrink temperature, °C (minimum)	98±0.5
Veslic rubbing abrasion, rating after 80 cycles (minimum) 2/	4, no damage

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TABLE I. Leather characteristics and requirements – Continued.

Characteristic	Requirement
Cold crack resistance, visual after exposure at -30 °C	No cracking
Finish adhesion, newtons/10 millimeters, dry and wet (minimum)	4
Flexibility and adhesion of finish at 30,000 cycles, at -20 °F	No cracking or delamination
Hexane soluble materials, percent (maximum) 3/	25
Chromic oxide, percent (minimum) 3/	3.0
pH value (minimum)	3.3
Total ash, percent (maximum) 3/	9.0
Determination of chromium (VI) content, mg/kg	<3

FOOTNOTES:

1/ Passing values in 80 percent of tested samples required to pass.

2/ Value given using the AATCC EP1.

3/ Calculated on moisture-free basis.

3.5.3 Grip, fingertips. Grip material shall be sewn at the fingertips on the outside of the liner to aid in maintaining grip when the liner is used alone as well as in combination with a shell. The material shall be a 100-denier polyester with a raised hexagonal structure made of silicone and shall conform to the requirements listed in table II when tested as specified in 4.4.3.

TABLE II. Grip requirements.

Characteristic	Requirement
Weight, ounces per square yard	11.0 to 14.0
Air permeability, cubic feet per minute (minimum)	18
Thickness (inch)	0.030 to 0.035
Fabric count, yarns per inch (minimum):	
Warp	125
Fill	100
Puncture propagation, kilogram-force (minimum):	
Warp	5
Fill	4
Abrasion, cycles (minimum)	3,000
Hydrostatic, centimeters (minimum)	18

3.5.4 Lining, inside cuff hem and trigger finger. The material used to line the inside cuff and trigger finger shall conform to MIL-DTL-32697.

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3.5.5 Lining, front of hand, back of hand, and thumb. The lining used on the front of the hand, the back of the hand, and the inner and outer thumb, shall be a napped lay-in knit fabric. The lay-in for the fabric shall be 100 percent wool and the knit back shall be a 100 percent recycled polyester. The 100 percent wool lay-in shall make up 70±2 percent of the total weight of the fabric and the recycled polyester back shall make up 30±2 percent of the total fabric weight. The wool fiber shall be carded and spun into single yarns. One end of wool yarn shall be laid in per knit course. After knitting, the fabric shall be scoured and napped to form a disoriented pile on the face to meet a minimum thickness of 0.090 inch measured under 0.1 pound per square inch (psi). The shade of the fabric shall be natural. The napped lay-in shall conform to the requirements listed in table III when tested as specified in 4.4.3.

TABLE III. Napped wool lay-in requirements.

Characteristic	Requirement
Fiber content	100% wool lay-in, 100% recycled polyester back
Weight, ounces per square yard	15.0 to 18.0
Thickness, inches, under 0.1 psi	0.090 to 0.1
Grade (minimum)	54s
Burst strength, pounds (minimum)	100
MVTR, grams per square meter in 24 hours, upright (minimum)	1,400
Dimensional stability, percent (maximum) (after five laundering cycles):	
Wales	-8/+1
Courses	-1/+8

3.5.6 Insulation. The insulation (see 6.6.4) used between the shell and the lining fabric on the front of the hand, the back of the hand, and the outer thumb shall conform to MIL-PRF-32567 type II, except the weight shall be 4.4±0.5 ounces per square yard; the minimal thickness at 0.01 psi shall initially be 0.60 inch and shall be 0.45 inch after three launderings; and the thermal resistance, in intrinsic clo, shall initially be 2.25 and shall be 1.95 after three launderings. Alternatively, the insulation may conform to MIL-PRF-32567 type III, class 4. For both types, a polyester blend may be used in lieu of the 100 percent polyester specified in MIL-PRF-32567.

3.5.6.1 Scrim. The spun bonded scrim specified in MIL-PRF-32567 shall be applied to one surface of any batting type used and shall be oriented during end item fabrication so the scrim is adjacent to the woven shell or knitted fabric liner, which can abrade the batting fibers causing clumping, roping, and pilling. The scrim shall have a nominal weight of 0.4 ounce per square yard and is additional weight over the table I batting weight for each type and class of MIL-PRF-32567. The scrim shall maintain its structural integrity when laundered as specified in MIL-PRF-32567. The scrim may be lightly adhered to batting if required for the end item construction.

3.5.7 Cuff. A cuff (see 6.6.6) shall be sewn onto the liner and provide a snug fit around the wrist measuring approximately 3 inches at the bottom edge of the cuff, when relaxed, for all sizes. The cuff shall conform to type I, class 2, size 2, of MIL-C-3735, except that it shall be 100 percent domestic wool at a minimum grade not coarser than US 60s to 62s (25 microns or finer) for comfort. Unless otherwise specified (see 6.2), the shade shall be Coyote 498 (see 6.5.2).

3.5.8 Thread. The thread for all construction seams shall be size letter B (Tex size 45) conforming to A-A-59826, type I, class B. Unless otherwise specified (see 6.2), the color shall be Coyote 498 (see 6.5.2) to approximate the mitten liner.

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3.5.9 Labels. Each liner shall have a combination identification-size label and instruction label. The inscription shall have a minimum font size of 10 points. The inscription legibility, label, and label attachment method shall last the expected life of the liner. The fastness to laundering requirements of MIL-DTL-32075 shall apply. The labels shall identify the item, boldly specify the size, and provide use and care instructions.

3.5.9.1 Identification-size label. The label shall be located on the outside of the terry lining, caught in the side seam of both mittens liners in a pair. The combination identification-size label shall conform to type VI, class 4, MIL-DTL-32075 and shall bear the following inscription:

LINER, TRIGGER FINGER MITTEN

SIZE: MEDIUM (EXAMPLE)

CONTRACT NO.:

CONTRACTOR'S NAME:

FIBER CONTENT:

3.5.9.2 Instruction label. The instruction label for the mitten liner shall conform to type VI, class 3 of MIL-DTL-32075, and shall include the following information:

1. Machine wash on regular cycle in cold water.
2. Do not bleach.
3. Drip dry.
4. Do not iron.
5. Do not dryclean.

DO NOT REMOVE THIS LABEL

3.5.9.3 Combination identification-size instruction label. As an alternate, a combination identification-size instruction label may be used conforming to type VI, class 14 of MIL-DTL-32075.

3.5.9.4 Label/tag. Each mitten liner shall have an individual barcode placed on a paper tag for personal clothing items conforming to type VIII, class 17 of MIL-DTL-32075. The paper tag shall contain the National Stock Number (NSN) and shall be standard bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tags shall have a hole and shall be attached to each pair of liners by a fastener. The tags shall be clearly legible and readable by a scanner.

3.6 Patterns. Standard patterns, providing a seam allowance of 1/8 inch for all seams, and 3/4 inch for the hem, will be furnished by the Government. The list of pattern parts and computer nomenclature in table IV is provided to ensure that the pattern set provided is complete. The Government patterns shall not be altered in any way, and are to be used only as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed, to facilitate manufacture, is permitted. Also, minor modifications are permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment. These modifications shall not alter the serviceability or appearance requirements of the liner.

3.6.1 List of pattern parts. The component parts shall be cut from the specified material in accordance with the pattern parts listed in table IV.

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TABLE IV. Pattern parts.

Material	Nomenclature	Cut Parts 1/
Liner shell	BACK OF HAND RIPSTOP	1 left, 1 right
	PALM RIPSTOP	1 left, 1 right
	OUTER THUMB RIPSTOP	1 left, 1 right
	INNER THUMB RIPSTOP	1 left, 1 right
	MAIN GUSSET RIPSTOP	1 left, 1 right
	PINKY GUSSET RIPSTOP	1 left, 1 right
	INSIDE THUMB GUSSET RIPSTOP	1 left, 1 right
	OUTSIDE THUMB GUSSET RIPSTOP	1 left, 1 right
Leather	TRIGGER FINGER BACK LEATHER	1 left, 1 right
	TRIGGER FINGER PALM LEATHER	1 left, 1 right
Grip	FINGERTIP GRIP	1 left, 1 right
Lining, napped wool lay-in	BACK OF HAND LINING	1 left, 1 right
	PALM LINING	1 left, 1 right
	OUTER THUMB LINING	1 left, 1 right
	INNER THUMB LINING	1 left, 1 right
Insulation	BACK OF HAND INSULATION	1 left, 1 right
	OUTER THUMB INSULATION	1 left, 1 right
	PALM INSULATION	1 left, 1 right
	INNER THUMB INSULATION	1 left, 1 right
Lining, wool terry	TRIGGER FINGER BACK TERRY	1 left, 1 right
	TRIGGER FINGER PALM TERRY	1 left, 1 right
	TRIGGER FINGER GUSSET TERRY	1 left, 1 right
	CUFF PALM TERRY	1 left, 1 right
	CUFF BACK TERRY	1 left, 1 right
Rib knit	Cuff	2

FOOTNOTE:

1/ With the exception of the cuffs, pieces shall be cut face-to-face to create left and right cut parts.

2/ Leather shall be cut in a manner that causes the appropriate side to be on the outside of the liner after assembly. This may require pieces to be flipped in the cutting software.

3.7 Construction. The assembled mitten liner (referred to as “end item” in this specification) shall be as specified in table V to ensure uniform appearance, comfort, and durability in garrison use, combat, and field operations. The end item construction and appearance shall conform to the requirements of this document, characteristics specified in table V, and the finished dimensions specified (see 3.9) to maintain item configuration and compliance to end item and component performance tests (see 4.4 and 4.5).

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TABLE V. Liner requirements.

Characteristic	Requirement	Test Method
Clo value (dry thermal insulation, initial) (minimum)	1.7	4.5.2
Clo value (dry thermal insulation, after 3X laundering) (minimum)	1.5	4.5.2 1/
FOOTNOTE: 1/ Launder only; omit dimensional changes.		

3.7.1 Stitches, seams, and stitching. All seams shall exhibit a uniform appearance and conform to ASTM D6193. Stitch types and stitches per inch shall be as listed in table VI. All material edges shall not ravel. Edges may be turned-in, turned-under, or serged to prevent raveling.

TABLE VI. Stitch types and stitches per inch.

Operation	Stitch Type	Stitches per Inch
Topstitch, single needle stitch, or straight stitch	301	8 to 12
Baste	301 or 101	8 to 12

3.7.2 Tacking and backstitching. Ends of seams and rows of stitching with 301 type stitch, when not caught in other seams or stitching, shall be securely backstitched not less than 1/4 inch. Thread breaks (all stitch types) shall be secured by stitching back from the break not less than 1/2 inch.

3.8 Color matching.

3.8.1 Visual shade matching. Unless otherwise specified (see 6.2), samples shall be verified for shade matching (see 6.5.2) in accordance with 4.5.3. The color and appearance of the cloth shall approximate the standard sample when tested as specified in 4.5.3.

3.8.2 Color. Unless otherwise specified (see 6.2), the exterior mitten liner materials shall be solid and shall be Coyote 498 in shade (see 6.5.2). The exterior materials and components shall meet the reflectance requirements specified in 3.8.2.1.

3.8.2.1 Spectral reflectance. The finished outer shell cloth and all of the components used shall meet the spectral reflectance values (in percent) for the visible/near infrared wavelength range, 600 to 860 nanometers for the colors specified in table VII as applicable when tested as specified in 4.5.4.

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TABLE VII. Spectral reflectance.

Wavelengths	Coyote 498	
Nanometers	Minimum	Maximum
600	8	20
620	8	20
640	8	22
660	8	24
680	12	24
700	12	34
720	16	42
740	22	46
760	30	50
780	34	54
800	36	56
820	38	58
840	38	58
860	40	60

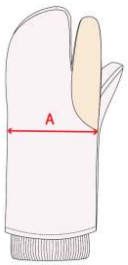
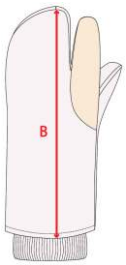
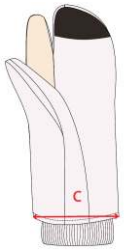
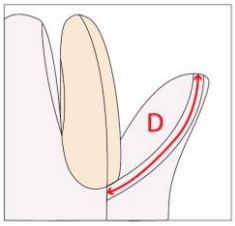
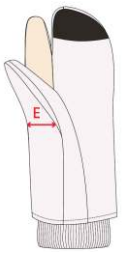
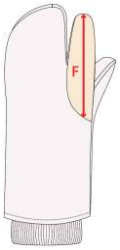
3.9 Finished measurements. Finished dimensions shall be in accordance with table VIII and measured in accordance with table IX.

TABLE VIII. Mitten liner finished dimensions (in inches).

Size	A: Back of Hand Width	B: Full Length	C: Wrist Cuff Width	D: Thumb Length	E: Thumb Width	F: First Finger Length	G: First Finger Width	H: Finger Mitt Width	I: Finger Mitt Length	J: Fourchette Width
Small	4-1/4	12-1/2	6	4	3	4-1/4	1-1/4	3-1/4	4	1/2
Medium	5	12-7/8	6-1/4	4-1/8	3	4-1/2	1-1/4	3-1/2	4-1/4	1/2
Large	5-1/4	13-1/4	6-1/2	4-1/4	3	4-3/4	1-1/2	3-3/4	4-1/2	1/2
Extra Large	5-1/2	13-5/8	6-3/4	4-3/8	3	5	1-1/2	4	4-3/4	1/2
Tolerance	±1/4	±1/4	±1/4	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8	±1/8

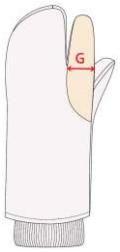
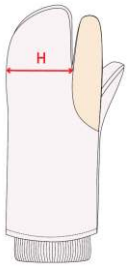
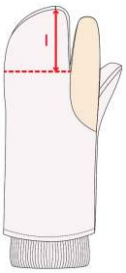
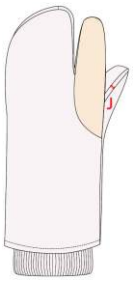
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TABLE IX. Mitten liner methods of measurement.

Measurement Name	Measurement Description	Figure
A: Back of hand width	Measurement taken on the back of the hand. Measurement parallel to the mitten liner opening, beginning at the side seam where the leather meets the nylon and ending at the opposite side seam.	
B: Full length	Measurement taken on the back of the hand beginning at the seam where the fourchette pieces meet to the bottom of the liner hem staying perpendicular to the liner hem.	
C: Wrist cuff width	Measurement taken on the palm side along the bottom edge of the cuff.	
D: Thumb length	Measurement taken on the back of the hand from the thumb tip following along the fourchette and ending where the fourchette meets the side seam.	
E: Thumb width	Measurement taken across the back of the thumb from seam to seam at the thumb crotch staying parallel with the thumb tip.	
F: First finger length	Measurement taken on the back of the hand at the center of the finger beginning at the seam where the leather meets the shell down to the bottom of the leather where it meets the nylon.	

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TABLE IX. Mitten liner methods of measurement – Continued.

Measurement Name	Measurement Description	Figure
G: First finger width	Measurement taken across the back of the first finger, from seam to seam, parallel to the fingertip, at the finger crotch.	
H: Finger mitt width	Measurement taken across the back of the finger mitt width, from seam to seam, beginning at the finger crotch to the pinky side seam, parallel to the fingertip.	
I: Finger mitt length	Measurement taken on the back of the hand, beginning at the seam where the fourchette pieces meet, to the finger crotch, staying perpendicular to the liner hem.	
J: Fourchette width	Measurement taken at the back of the hand across the fourchette, 1 inch up from the thumb crotch seam, staying parallel to the tip of the thumb.	

3.10 Workmanship. The finished mitten liners shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished liners shall be uniform in quality, free from defects that adversely affect form, fit, or function, and the defects specified in 4.4.1. The occurrence of defects shall not exceed the established acceptance quality limits (AQL) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. A first article inspection shall consist of the examinations and tests specified in 4.4. The first article shall be rejected if any associated requirements are not met.

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4.2.1 First article samples. The sample size will be specified in the procurement document (see 6.2). The sample unit shall be one pair of mitten liners and the lot size shall be expressed in pairs.

4.3 Conformance inspection. Conformance inspection shall include the examination specified in 4.3.1.

4.3.1 Conformance inspection samples. Sampling for conformance inspection shall be performed in accordance with ASQ/ANSI Z1.4 and with AQLs as specified in the contract or order (see 6.2). The sample unit shall be one pair of mitten liners and the lot size shall be expressed in pairs.

4.3.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether the construction details that cannot be easily examined in the finished product are in accordance with requirements specified in section 3. Materials and components that contain a defect in accordance with 4.4.1 shall be corrected or removed from production.

4.4 Component and end item inspection. In addition to testing provisions specified in reference documents, components shall be tested for the characteristics specified in this specification. Tests shall be conducted with both the specimen and test apparatus under standard conditions as defined in the various ASTM and AATCC test methods. All requirements are applicable to the sample unit. Test reports, when prescribed in the associated test method, evaluation procedure or other cited guidance, should contain the individual values utilized in expressing final results. The lot shall be rejected if one or more sample units fail to meet any test requirement specified.

4.4.1 End item visual examination. The mitten liner shall be examined for defects in shade, design, material, construction, and workmanship, with defects classified in accordance with table X. Defects shall not exceed the AQL specified in the contract or order (see 6.2).

TABLE X. Defects.

Examination	Defect	Major	Minor
Mitten pairing	Not matched and paired left and right, right and left not same size, wide variation in appearance	101	
	Not tacked or joined together as specified.		201
Design	Not as specified (incorrect material, pattern, etc.)	102	
	Poorly shaped		202
Shade	Shade variation within a part or between parts	103	
	Unacceptable shade change as a result of water repellent treatment application	104	
	Any color not as specified	105	
Cleanness	Spot, stain		203
	Excessive thread ends not trimmed or removed		204
	Shade stamp marking on outside		205
	Odor, affecting appearance or serviceability		206

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TABLE X. Defects – Continued.

Examination	Defect	Major	Minor
Material and workmanship	Any component part omitted, distorted, full, tight, or twisted	106	
	Not functioning properly or defective	107	
	Any part caught in any unrelated stitching	108	
	Cloth material:		
	Any hole, cut, tear, smash, burn, drill hole, run, thin place, dye streak, misweave or misknit, knot or slub affecting appearance or serviceability	109	
	Leather:		
	Not specified color	110	
	Not full grain	111	
	Flesh side not smooth or contains area of coarse, loose fibers	112	
	Not clean–stained, or foreign matter		207
	Hard, boney, loose, spongy leather, hard scar	113	
	Cut or hole	114	
	Seam (defect affecting appearance or serviceability)		
	Not specified type	115	
	Puckered, distorted, pleated, wavy, twisted, irregular, or open	116	
	Loose or tight stitch tension, broken or missing thread or stitch	117	
	Needle cutting, needle chew, visible mend	118	
	Edge or raised stitching sewn too close to the edge resulting in damage to the cloth	119	
	Seam allowance not as specified	120	
	Visible raw edge	121	
	Any visible perforations from removed stitches		208
Material and workmanship (continued)	Poorly assembled and affecting serviceability	122	
	Difference in overall length between the front and back of the mitten liner is greater than 1/2 inch	123	
	Unwanted permanent fold, pleat, or crease in the fabric or mitten, affecting appearance or serviceability	124	
Labels	Omitted, incorrect, illegible, not attached where specified		209
	Barcodes omitted, not readable by scanner		210
	Human-readable interpretation omitted or illegible		211
	Barcode not visible on folded, packaged item; barcode attachment causes damage to the item.		212
Packaging	Not packaged in accordance with the contract or purchase order (see 6.2)		213

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4.4.2 Dimensional examination. The end item shall be examined for conformance to the dimensions specified in table VIII.

4.4.3 Component testing. The components (see 3.5.2 through 3.5.5) shall be tested for characteristics listed in table XI. The methods of test shall be as specified in table XI. The testing requirements and test sampling plan shall be as specified in the test method.

TABLE XI. Component tests.

Characteristic	Requirement Paragraph	Test Method
Thickness	Table I	ASTM D1814
MVTR	Table I	ASTM D5052
Water absorption resistance	Table I	4.5.1
Oil and stain resistance	Table I	AATCC TM118
Abrasion resistance	Table I	ASTM D7255
Stiffness	Table I	ASTM D2821
Stitch tear	Table I	ASTM D4705
Elongation at 25 pound load	Table I	ASTM D2209
Shrink temperature	Table I	ASTM D6076
Veslic rubbing abrasion 1/	Table I	ISO 11640
Cold crack resistance	Table I	ASTM D1912
Finish adhesion	Table I	ASTM D6182
Flexibility and adhesion of finish at 30,000 cycles	Table I	ASTM D6182
Hexane soluble material 2/	Table I	ASTM D3495
Chromic oxide 2/	Table I	ASTM D2807
pH value	Table I	ASTM D2810
Total ash 2/	Table I	ASTM D2617
Determination of chromium (VI) content	Table I	ISO 17075-2
Weight	Tables II and III	ASTM D3776/D3776M
Air permeability	Table II	ASTM D737
Thickness	Table II	ASTM D1777
Fabric count	Table II	ASTM D3775
Puncture propagation	Table II	ASTM D2582
Abrasion	Table II	ASTM D3884
Hydrostatic pressure	Table II	AATCC TM127
Fiber content 3/	Table III	AATCC TM20
Thickness	Table III	ASTM D1777
Grade 4/	Table III	USDA United States Standards for Grades of Wool
Burst strength	Table III	ASTM D6797
MVTR	Table III	ASTM E96/E96M, procedure B
Dimensional stability	Table III	AATCC TM135, 2, II, Aii

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TABLE XI. Component tests – Continued.

FOOTNOTES:

1/ Value shall be obtained using the AATCC EP1.

2/ Testing shall be conducted and result calculated on moisture-free basis.

3/ Testing shall be conducted on a scoured form.

4/ A certificate of compliance may be submitted to address the stated requirement (see 6.7).

4.4.4 End item testing. The finished liners shall be tested for the characteristics listed in table XII according to the specified methods. The testing requirement and test sampling plan shall be specified in 4.5.2.

TABLE XII. End item testing.

Characteristic	Requirement Paragraph	Verification
Clo (dry thermal insulation) value	3.7	4.5.2

4.5 Examinations and tests.

4.5.1 Water absorption resistance test. One 3-inch diameter specimen shall be tested from each 12-by-12-inch test area. Prior to testing, the specimen shall be allowed to reach moisture equilibrium in the standard atmospheric conditions environment, as specified in ASTM D1610. A stainless steel 500-milliliter, 3-inch diameter, 5-inch-long container with a rubber gasket and cover shall be used to test the specimen. The specimen shall first be weighed (to the nearest 0.01 gram) and then shall be placed inside the cover with the flesh side of the specimen next to the inside of the cover. The rubber gasket shall then be placed inside the cover and against the grain surface of the leather specimen. One-hundred milliliters of distilled or deionized water at 23 °C shall then be added to the container. The cover with the specimen and gasket shall then be secured to the container. The container shall then be inverted so the water is against the grain surface of the specimen. The inverted container shall be placed on a flat surface and allowed to remain in the inverted position for 30 minutes. At the end of 30 minutes, the container shall be placed upright and the cover shall be removed and placed on a flat surface. The inside of the cover shall face the flat surface. The cover shall be tilted so the outer edge of the bottom is raised 1 to 1-1/2 inches from the flat surface and allowed to drain for 5 minutes. After 5 minutes, the specimen shall be removed from the cover, held by its edge, and shook briskly once to remove any free moisture from the surface of the specimen. The specimen shall be immediately reweighed to the nearest 0.01 gram; the percent water absorption shall be calculated as follows:

$$\text{Percent water absorption} = \frac{(W2 - W1) \times 100}{W1}$$

Where:

W1 = Original weight of conditioned specimen

W2 = Weight of specimen after being subject to water

4.5.2 Thermal insulation test. Three separate samples of the mitten liners shall be evaluated in accordance with ASTM F3426 and the average value is reported. Total thermal resistance (R) values can be converted to the more familiar clo unit, by multiplying the R value by 6.45, to allow ranking order of standard for procurement purposes. Test size shall be right hand, size large. A life-sized biophysical model of the human hand that measures localized and total thermal resistance shall be used. Insulation values for the total hand model and its seven individual regions are calculated by an internal program during 30 minutes of steady state data collection. The ambient temperature shall be maintained at 5±0.5 °C and the surface of the hand model is controlled at 35±0.5 °C. The relative humidity shall be 50±5 percent and wind speed shall be maintained at 0.4±0.1 meter per second. Samples evaluated after laundering shall be laundered in accordance with AATCC TM150, option 1, III, A, iii. Total thermal resistance (R) to heat exchange is calculated in the test conditions stated above and using:

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$$R = \frac{(SA \times \text{delta T})}{P}$$

Where:

R is the thermal resistance, m^2 °C/watt

SA = total surface area of seven regional segments, m^2

Delta T = temperature gradient between the hand model surface and ambient air temperature, °C

P = Total power input, watts

If required (see 6.2), mitten liners submitted will be tested by the Government. Testing at alternate facilities shall require side-by-side comparison due to chamber wind speed differences that cannot be controlled that will affect the total measured values.

4.5.3 Visual shade matching. Samples shall be viewed using AATCC EP9, option C (see 6.5), with sources simulating artificial daylight D65 illuminant (see 6.5.1) with a color temperature of 6500±200 Kelvin (K) illumination of 100±20 foot candles, and under incandescent A illuminant with a color temperature of 2856±200K.

4.5.4 Spectral reflectance test. Spectral reflectance data shall be obtained from 600 to 860 nanometers at 20-nanometer intervals on a spectrophotometer relative to the polytetrafluoroethylene family of compounds, the preferred white standard. Other white reference materials may be used provided they are calibrated to absolute white or vitrolite tiles. The spectral band width shall be less than 20 nanometers at 860 nanometers. Reflectance measurements shall be made by either the monochromatic or polychromatic mode of operation. When the polychromatic mode of operation is used, the spectrophotometer shall operate with the specimen diffusely illuminated with the full emission of a continuous source that simulates in the visible spectrum either International Commission on Illumination (CIE) source A or CIE source D65 (see 6.5.1). The specimen shall be viewed at an angle of no greater than 10 degrees from normal, with the specular component included. Measurements shall be taken on a minimum of two different areas and the data averaged. The specimen shall be measured as a single layer backed with three layers of the same fabric and shade. Specimens shall be oriented in different directions during testing. When possible, the specimens tested shall not contain the same warp and filling yarns. The diameter for standard aperture size used in the color measurement device shall be the largest possible for each color. The measured areas should be at least 6 inches away from the selvage. Photometric accuracy of the spectrophotometer shall be within 1 percent and wavelength accuracy within 2 nanometers. Any color having spectral reflectance values falling outside the limits at four or more of the wavelengths specified shall be considered a test failure.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This liner is intended for wear by military personnel of the Department of Defense during extreme cold weather conditions in combination with a full mitten shell or a trigger finger mitten shell conforming to types I and II of MIL-DTL-32707.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sizes required (see 1.2).

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c. The specific issue of individual documents referenced (see 2.2 and 2.3).

d. Applicable Government patterns, including revisions.

e. When first article sample is required (see 3.1.1, 4.2, and 6.3).

f. When conformance inspection is required (see 3.1.2).

g. Shade standard information (see 3.5.2, 3.5.7, 3.5.8, 3.8.1, 3.8.2, and 6.5.2).

h. AQLs (see 3.10, 4.2.1, and 4.4.1).

i. Number of first article inspection samples (see 4.2.1).

j. When the mitten liners require thermal insulation testing by the Government (see 4.5.2).

k. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product that has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

6.4 Standard sample. For standard samples, address the procuring activity issuing the invitation for bids or request for proposal.

6.5 Visual shade matching. In 2017, option A of AATCC EP 9 was changed to option C. NOTE: In case of confusion, the viewing geometry should be such that the specimen plane and illumination source are parallel to each other and aligned so that the light flux is incident at the center of the specimen plane, which is set at a 35±5-degree angle relative to the horizontal. The observer will view the specimens at a 90-degree angle, relative to the plane of the specimens.

6.5.1 D75 illuminant. The use of D75 illuminant with a color temperature of 7500±200K and illumination of 100±20 foot candles in lieu of the specified D65 illuminant (see 4.5.4) is permitted.

6.5.2 Shade criticality. Some items may be deemed "non-shade-critical" by the contracting agency and alternative shade standards or information regarding shade may be specified in the contract or order (see 6.2). It is recommended that manufacturers refer to the contract or order to determine the criticality of the shade matching or alternate shade standards that are acceptable.

6.6 Sources of supply.

6.6.1 Basic shell material. 70 denier nylon ripstop (see 3.5.1) has been shown to meet the requirements of this specification. A known source of this material is:

Brookwood Companies, Inc.

485 Madison Avenue, Suite 500

New York, NY 10022

Phone: 212-551-0100

Fax: 646-472-0294

https://brookwoodcompanies.com/

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6.6.2 Grip, fingertips. Style #700554-700 (see 3.5.3) has been shown to meet the requirements of this specification. A known source of this material is:

SuperFabric
570 Hale Ave. N.
Oakdale, MN 55128
Phone: 651-730-6203
www.superfabric.com

6.6.3 Lining, inside cuff hem and trigger finger. 100 percent wool napped French terry, style # 2196-10 (see 3.5.4) has been shown to meet the requirements of this specification. A known source of this material is:

Draper Knitting Company
28 Draper Lane
Canton, MA 02021-1693
Phone: 1-800-808-7707
www.draperknitting.com

6.6.4 Lining, front of hand, back of hand, and thumb. Napped wool lay-in, style #2162 (see 3.5.5) has been shown to meet the requirements of this specification. A known source of this material is:

Draper Knitting Company
28 Draper Lane
Canton, MA 02021-1693
Phone: 1-800-808-7707
www.draperknitting.com

6.6.5 Insulation. A known source that meets the requirements of the material specified in 3.5.6 is:

Climashield
641 Northpark Drive
Clinton, TN 37716
Phone: 865-463-6184
Fax: 865-463-6188
www.climashield.com

6.6.6 Cuff, rib. A known source that meets the requirements of the material specified in 3.5.7 is:

Minnesota Knitting Mills
1450 Mendota Heights Rd
St. Paul, MN 55120
Phone: 651-452-2240
sales@mnknit.com

6.7 Certificates of compliance. A certificate of compliance may be acceptable for addressing selected test requirements at the discretion of the contracting agency. It is recommended that contractors submit a request to the contracting agency to obtain permission to use certificates of compliance before relying on their acceptance.

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6.8 NSN information. The NSN for a representative sample of the mitten liners is provided below. This is not intended to be an exhaustive list.

Size	NSN
Medium	8415-01-697-9022

6.9 Subject term (key word) listing.

Handwear

Leather

Nylon

Outer shell

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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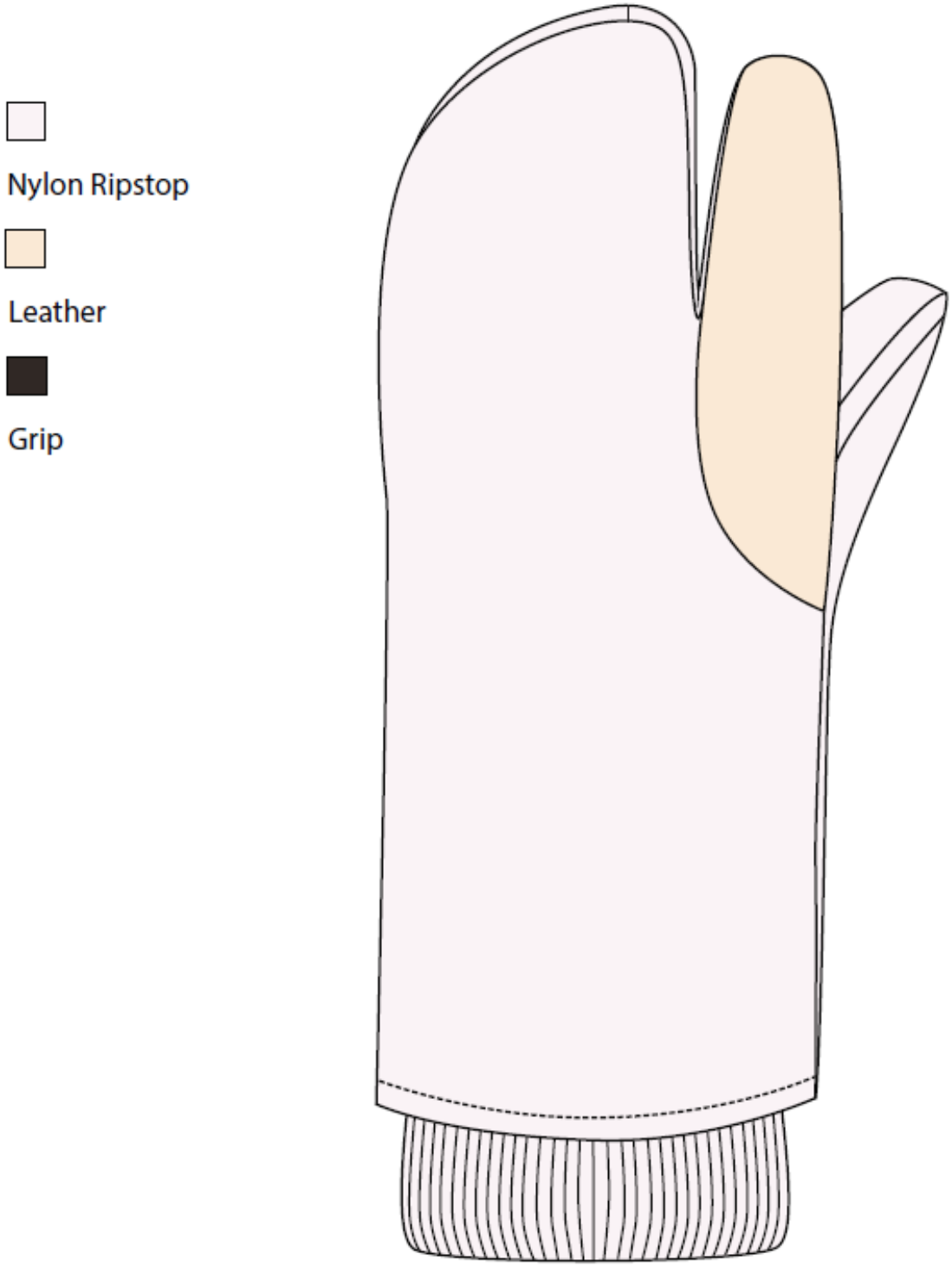


FIGURE 1. Back of hand view.

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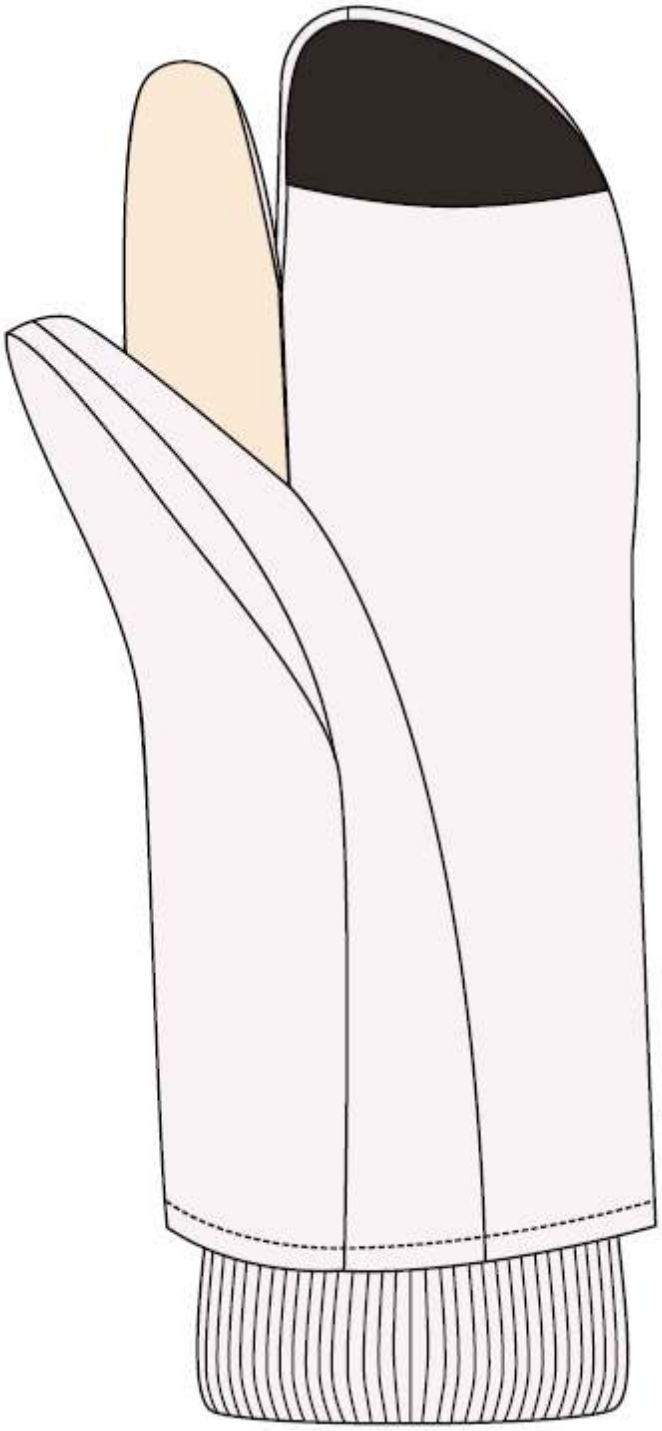


FIGURE 2. Palm view.

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CONCLUDING MATERIAL

Custodians:
Army – GL
Navy – MC

Review activities:
Army – AV, MI
Navy – AS, CG1, NU
Air Force – 11
DLA – CT

Civil agency:
GSA – FAS

Preparing activity:
Navy − MC
(Project 8440-2024-002)

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